holds for all $i \in [n]$ and $t \geqslant 0$.

In fact, let Q and K be arbitrary $d \times d$ matrices. Then the same result also holds for the solution to the corresponding Cauchy problem for (2.3) with $V = I_d$ (or the natural analogue of (USA) with these parameters).

Remark 6.5. *Lemma 6.4 implies that $\left\{(\bar{x}_i)_{i\in[n]} \in (\mathbb{S}^{d-1})^n : \bar{x}_1 = \ldots = \bar{x}_n\right\}$ is Lyapunov asymptotically stable as a set. In fact, it is exponentially stable.*

Lemma 6.4 is reminiscent of results on interacting particle systems on the sphere (see [CLP15, Theorem 3.7] for instance), and the literature on synchronization for the Kuramoto model on the circle ([ABK+22, Lemma 2.8], [HR20, Theorem 3.1] and Section 7.2). We often make use of the following elementary lemma.

Lemma 6.6. *Let $f : \mathbb{R}_{\geqslant 0} \to \mathbb{R}$ be a differentiable function such that*

$$\int_0^{+\infty} |f(t)| \, \mathrm{d}t + \sup_{t\in\mathbb{R}_{\geqslant 0}} \left|\dot{f}(t)\right| < +\infty.$$

Then $\lim_{t\to+\infty} f(t) = 0$.

The proof of Lemma 6.4 is an adaptation of [CLP15, Theorem 1]. We present it here for completeness.

Proof of Lemma 6.4. We focus on the case (USA), and set

$$a_{ij}(t) := n^{-1} e^{\beta\langle x_i(t), x_j(t)\rangle} > 0.$$

The proof for (SA) is identical, and one only needs to change the coefficients $a_{ij}(t)$ by $Z_{\beta,i}(t)^{-1} e^{\beta\langle x_i(t), x_j(t)\rangle}$ throughout. Also note that since we only make use of the positivity of the coefficients $a_{i,j}(t)$ throughout the proof, all arguments are readily generalizable to the case of arbitrary $d \times d$ matrices Q and K appearing in the inner products.

Step 1. Clustering. For $t \geqslant 0$, consider

$$i(t) \in \operatorname*{arg\,min}_{i\in[n]} \langle x_i(t), w\rangle.$$

Fix $t_0 \geqslant 0$. We have

$$\begin{aligned}
&\left(\frac{\mathrm{d}}{\mathrm{d}t}\langle x_{i(t_0)}(\cdot), w\rangle\right)\Big|_{t=t_0} \\
&= \sum_{j=1}^{n} a_{i(t_0)j}(t_0)\Big(\langle x_j(t_0), w\rangle - \langle x_{i(t_0)}(t_0), x_j(t_0)\rangle\langle x_{i(t_0)}(t_0), w\rangle\Big) \geqslant 0.
\end{aligned}$$

This implies that all points remain within the same open hemisphere at all times and the map

$$t \mapsto r(t) := \min_{i\in[n]} \langle x_i(t), w\rangle$$

is non-decreasing on $\mathbb{R}_{\geqslant 0}$. It is also bounded from above by 1. We may thus define $r_\infty := \lim_{t\to+\infty} r(t)$. Note that $r_\infty \geqslant r(0) > 0$ by assumption. By compactness, there exist a sequence of times $\{t_k\}_{k=1}^{+\infty}$ with $t_k \to +\infty$, and some $(\overline{x}_i)_{i\in[n]} \in (\mathbb{S}^{d-1})^n$ such that $\lim_{k\to+\infty} x_i(t_k) = \overline{x}_i$ for all $i \in [n]$. Using the definition of $r(t)$, we also find that

$$\langle \overline{x}_j, w\rangle \geqslant r_\infty$$

for all $j \in [n]$, and by continuity, there exists $i \in [n]$ such that $\langle \overline{x}_i, w\rangle = r_\infty$. Then

(6.2)
$$\lim_{k\to+\infty} \langle \dot{x}_i(t_k), w\rangle = \sum_{j=1}^n \overline{a}_{ij}(\langle w, \overline{x}_j\rangle - \langle \overline{x}_i, \overline{x}_j\rangle\langle \overline{x}_i, w\rangle) \geqslant r_\infty \sum_{j=1}^n \overline{a}_{ij}(1 - \langle \overline{x}_i, \overline{x}_j\rangle),$$

where we set $\overline{a}_{ij} := e^{\beta\langle \overline{x}_i, \overline{x}_j\rangle} > 0$. Notice that

$$\lim_{k\to+\infty} \int_{t_k}^{+\infty} \langle \dot{x}_i(s), w\rangle \mathrm{d}s = r_\infty - \lim_{k\to+\infty} \langle x_i(t_k), w\rangle = 0,$$

and by using the equation (USA) we also find that $|\langle \ddot{x}_i(t), w\rangle| = O(e^{2\beta})$ for any $t \geqslant 0$. Therefore by Lemma 6.6, the left-hand side of (6.2) is equal to 0, and consequently the right-hand side term as well. This implies that $\overline{x}_1 = \ldots = \overline{x}_n := x^*$. Repeating the argument by replacing w with x^*, we see that the extraction of a sequence $\{t_k\}_{k=1}^{+\infty}$ as above is not necessary, and therefore

$$\lim_{t\to+\infty} x_i(t) = x^* \tag{6.3}$$

for all $i \in [n]$.

Step 2. Exponential rate. We now improve upon (6.3). Set

$$\alpha(t) := \min_{i\in[n]} \langle x_i(t), x^*\rangle.$$

From (6.3) we gather that there exists some $t_0 > 0$ such that $\alpha(t) \geqslant \frac{1}{2}$ for all $t \geqslant t_0$. Also, in view of what precedes we know that x^* lies in the convex cone generated by the points $x_1(t), \ldots, x_n(t)$ for any $t > 0$. Thus, there exists some $\eta \in (0, 1]$ such that ηx^* is a convex combination of the points $x_1(t), \ldots, x_n(t)$, which implies that

$$x^* = \sum_{k=1}^n \theta_k(t) x_k(t), \qquad \text{for some} \qquad \sum_{k=1}^n \theta_k(t) \geqslant 1, \quad \theta_k(t) \geqslant 0 \quad \forall k \in [n]. \tag{6.4}$$

For any t, we denote by $i(t)$ an element of $\arg\min(\langle x_i(t), x^*\rangle)$ for which $\langle \dot{x}_i(t), x^*\rangle$ is smallest. It follows from a Taylor expansion of $\langle x_i(t+h), x^*\rangle$ for $h > 0$ and $i \in [n]$ that

$$\dot{\alpha}(t) = \langle \dot{x}_{i(t)}(t), x^*\rangle.$$

Therefore

$$\dot{\alpha}(t) = \langle \dot{x}_{i(t)}(t), x^*\rangle \geqslant \sum_{j=1}^n a_{i(t)j}(t)(1 - \langle x_{i(t)}(t), x_j(t)\rangle)\alpha(t) \tag{6.5}$$

On another hand,

$$\min_{j\in[n]} \langle x_{i(t)}(t), x_j(t)\rangle \leqslant \sum_{k=1}^n \theta_k(t)\langle x_{i(t)}(t), x_k(t)\rangle = \langle x_{i(t)}(t), x^*\rangle = \alpha(t). \tag{6.6}$$

Plugging (6.6) into (6.5) and using $a_{ij}(t) \geqslant n^{-1}e^{-2\beta}$ we get

$$\dot{\alpha}(t) \geqslant \frac{1}{2ne^{2\beta}}(1 - \alpha(t)) \tag{6.7}$$

for $t \geqslant t_0$. Applying the Grönwall inequality we get

$$1 - \alpha(t) \leqslant \frac{1}{2} e^{-\frac{1}{2ne^\beta}(t-t_0)} \tag{6.8}$$

for all $t \geqslant t_0$. The conclusion follows. □

In the case $d < n$, we can still apply Wendel's theorem (recalled below) together with Lemma 6.4 to obtain clustering to a single point with probability at least $p_{n,d}$ for some explicit $p_{n,d} \in (0,1)$.

Theorem 6.7 (Wendel, [Wen62])**.** *Let $d, n \geqslant 1$ be such that $d \leqslant n$. Let $x_1, \dots, x_n$ be n i.i.d. uniformly distributed points on $\mathbb{S}^{d-1}$. The probability that these points all lie in the same hemisphere is:*

$$\mathbb{P}\Big(\exists w \in \mathbb{S}^{d-1} \colon \langle x_i, w\rangle > 0 \quad \text{for all} \quad i \in [n]\Big) = 2^{-(n-1)} \sum_{k=0}^{d-1} \binom{n-1}{k}.$$

6.2. **More precise quantitative convergence.** When n is fixed and $d \to +\infty$, in addition to showing the formation of a cluster as in Theorem 6.3, it is possible to quantitatively describe the entire evolution of the particles with high probability. To motivate this, on the one hand we note that since the dynamics evolve on $\mathbb{S}^{d-1}$, inner products are representative of the distance between points, and clustering occurs if $\langle x_i(t), x_j(t)\rangle \to 1$ for any $(i,j) \in [n]^2$ as $t \to +\infty$. On the other hand, if $d \gg n$, n points in a generic initial sequence are *almost orthogonal* by concentration of measure [Ver18, Chapter 3], and we are thus able to compare their evolution with that of an initial sequence of truly *orthogonal* ones.

We begin by describing the case of exactly orthogonal initial particles, which is particularly simple as the dynamics are described by a single parameter.

Theorem 6.8. *Let $\beta \geqslant 0$, $d, n \geqslant 2$ be arbitrary. Consider an initial sequence $(x_i(0))_{i\in[n]} \in (\mathbb{S}^{d-1})^n$ of n pairwise orthogonal points: $\langle x_i(0), x_j(0)\rangle = 0$ for $i \neq j$, and let $(x_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ denote the unique solution to the corresponding Cauchy problem for (SA) (resp. for (USA)). Then the angle $\angle(x_i(t), x_j(t))$ is the same for all distinct $i, j \in [n]$:*

$$\angle(x_i(t), x_j(t)) = \theta_\beta(t)$$

for $t \geqslant 0$ and some $\theta_\beta \in C^0(\mathbb{R}_{\geqslant 0}; \mathbb{T})$. Furthermore, for (SA), $\gamma_\beta(t) := \cos(\theta_\beta(t))$ satisfies

$$\begin{cases} \dot\gamma_\beta(t) = \dfrac{2e^{\beta\gamma_\beta(t)}(1-\gamma_\beta(t))((n-1)\gamma_\beta(t)+1)}{e^\beta + (n-1)e^{\beta\gamma_\beta(t)}} & \text{for } t \geqslant 0 \\ \gamma_\beta(0) = 0\,, \end{cases} \tag{6.9}$$

and for (USA), we have

$$\begin{cases} \dot\gamma_\beta(t) = \dfrac{2}{n} e^{\beta\gamma_\beta(t)}(1-\gamma_\beta(t))((n-1)\gamma_\beta(t)+1) & \text{for } t \geqslant 0 \\ \gamma_\beta(0) = 0\,. \end{cases} \tag{6.10}$$

Here and henceforth, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ denotes the one-dimensional torus. We provide a brief proof of Theorem 6.8 just below. The following result then shows that when $d \gg n$, $t \mapsto \gamma_\beta(t)$ is a valid approximation for $t \mapsto \langle x_i(t), x_j(t)\rangle$ for any distinct $i, j \in [n]$.

Theorem 6.9. *Fix $\beta \geqslant 0$ and $n \geqslant 2$. Then there exists some $d^*(n,\beta) \geqslant n$ such that for all $d \geqslant d^*(n,\beta)$, the following holds. Consider a sequence $(x_i(0))_{i\in[n]}$ of n i.i.d. uniformly distributed points on $\mathbb{S}^{d-1}$, and let $(x_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ denote the unique solution to the corresponding Cauchy problem for (SA). Then*